Markov Decision Process with finite time criteria

Finite Horizon Dynamic Programming for Hydro Valleys



The costs of exact dynamic programming, and an analysis of whether it fails safely

Christopher Won

Master 2 Optimization · École Polytechnique, Université Paris-Saclay

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The Problem

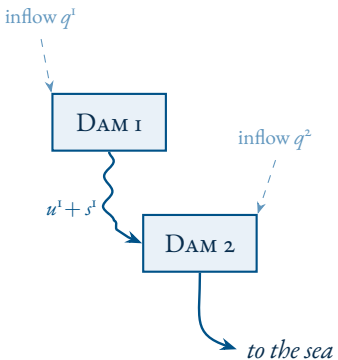


Plate I. *What dam 1 releases, dam 2 receives.*

- Each month, choose how much water each dam turbines.
- Inflows are random; water is sold at every dam it passes.
- State = volume in every dam. Bellman, backwards over 12 months:

$$\bar{v}_t(x) = \mathbb{E}\Big[\min_u \{\, c_t(x,u,W) + \bar{v}_{t+1}(f_t(x,u,W)) \,\}\Big]$$

The question of this talk

Exact DP is everyone's reference solution, but it only runs on a grid. *What does the grid cost, and does it fail safely?*

One Transition, Four Outcomes

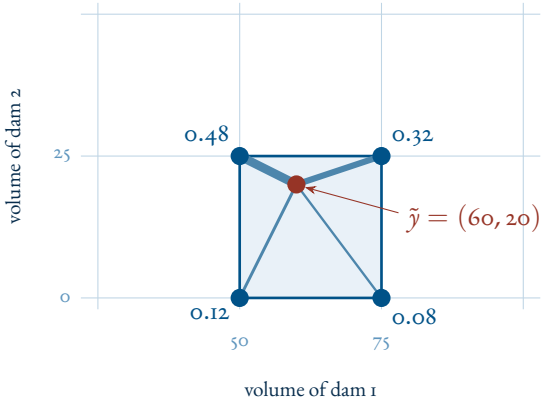


Plate II. *The next state falls between grid points.*

- Next month's volume falls *between* grid points.
- Rounding snaps it to one corner. We **split it** over all four.
- The weights are probabilities; their average is exactly the true point:

$$0.12\,(50, 0) + 0.48\,(50, 25) + 0.08\,(75, 0) + 0.32\,(75, 25) = (60, 20)$$

Not a patch on the model, but an honest MDP: exact for Bellman, and ***unbiased****.*

The Model That Over-Promises

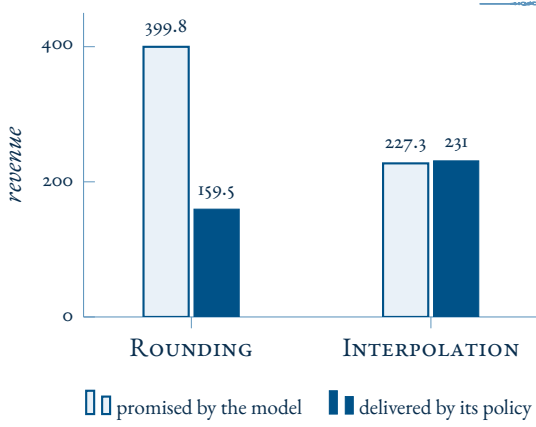


Plate III. *Two dams, coarsest grid ($n = 6$).*

- Rounding promises 2.5× the revenue its policy delivers.
- Interpolation very slightly **under**-promises, on every grid tested.

THE SAME ACCURACY NEEDS

121 *vs* 103 041

interpolation *rounding*

grid points, two dams

an 852-fold smaller state space

What the Policy Learned

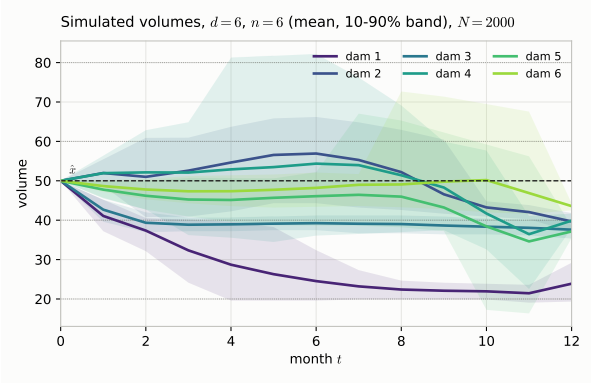


Plate IV. *Six dams, mean simulated volumes.*

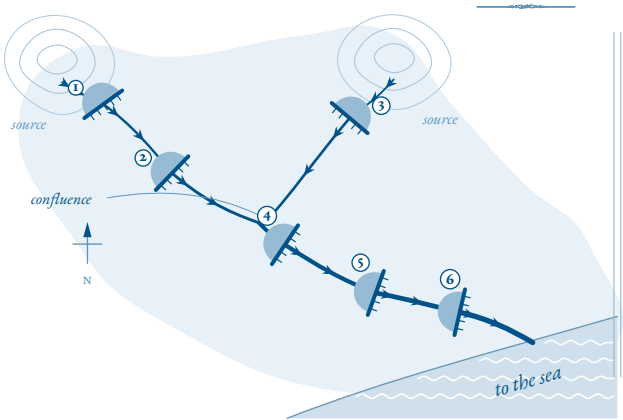
Given only costs and dynamics, it behaves like a water manager:

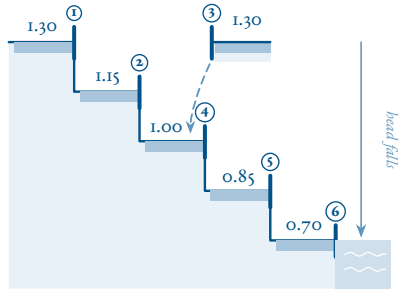
- **Drains the top dam first**: a unit released there is sold at five dams on the way out.

  5.00 *top of the valley* vs 0.70 *at the outlet*

- Stores through the wet spring, releases in autumn.

The Six-Dam Valley





$1.30 + 1.15 + 1.00 + 0.85 + 0.70 = 5.00$
for one unit held at the top

Plate V. *Six dams, two sources, one confluence: the valley from above, and the same dams by height.*

The Real Frontier

Six dams, exact backward induction

110 *minutes*, this work

10^8 *minutes*, extrapolated by Carpentier et al. (2018)

- They priced a *reference-grade* grid; we accept 7 points per dam.
- The limit is **the mesh you accept**, not the number of dams.

And what that mesh costs

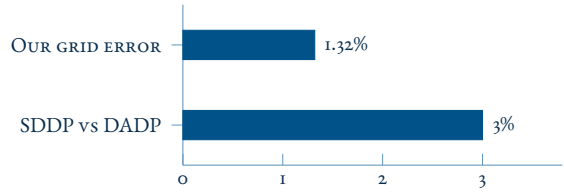


Plate V. *Six dams. Our grid error is the same order as the disagreement between the two decomposition methods.*

In Conclusion

I. Splitting each transition over the cell's corners makes the grid model an **honest, unbiased MDP**.

II. It is **852× cheaper** than rounding for the same accuracy, and it **errs on the safe side**.

III. The frontier of exact DP is **the resolution you accept**, not the number of dams.

Thank you

Appendix · The Bellman Recursion

Hazard–Decision: the month's inflow is seen before the release is chosen, so the expectation sits *outside* the minimum. State augmented to (X_t, W_t), then reduced to E:

$$\bar{v}_t(x) = \mathbb{E}\Big[\min_{u \in C_t(x, W_t)} \Big\{ c_t(x, u, W_t) + \sum_{\kappa \in \{0,1\}^d} \omega_\kappa\big(f_t(x, u, W_t)\big)\, \bar{v}_{t+1}(g_{k+\kappa}) \Big\}\Big], \qquad \bar{v}_T = \phi.$$

$$\omega_\kappa(y) = \prod_{i=1}^{d} \big(\kappa^i \lambda^i + (1-\kappa^i)(1-\lambda^i)\big), \qquad \sum_\kappa \omega_\kappa = 1, \qquad \sum_\kappa \omega_\kappa\, g_{k+\kappa} = y.$$

Cost of the backward pass: $\Theta\big(T\, L\, d\, (2nm)^d\big)$ (n^d states, m^d controls, 2^d corners). Rounding saves the 2^d, but the exponential in d remains.

Appendix · Does the Error Grow with d?

d	gap	relative	gap / d
2	6.01	2.68%	3.01 ± 0.28
4	11.12	1.48%	2.78 ± 0.44
6	20.73	1.35%	3.46 ± 0.58

Plate A. *Fixed mesh $\delta = 25$ and fixed $m = 4$; $N = 10^4$.*

- It *appeared* to grow: but the 6-dam runs also had a coarser release grid ($m = 4$ vs 10).
- With both grids held fixed, the relative error **falls**.
- Each reservoir adds roughly the same bias, ≈ 3; the biases add.

Appendix · The Policy Converges Before Its Value

d	mesh δ	predicted $\bar{v}_0(x_0)$	realised $\hat{J}_N$	gap
4	25.00	-746.51	-754.51	$+8.00$
4	14.29	-768.78	-774.58	$+5.80$
4	9.09	-772.81	-774.99	$+2.18$
4	6.67	-773.86	-775.42	$+1.56$

- From $\delta = 14.29$ on, the realised cost moves by 0.84 (0.11%); the prediction still moves by 5.08.
- A coarse grid gives a **near-optimal policy** with an **unreliable price tag**.
- Hence the six-dam value is reported as *at most* -1538.